

Project Proposal for K-12 School District Enrollment Projections

Executive Summary

This proposal outlines a plan to enhance the K-12 school district's enrollment projection process by implementing a machine learning (ML) model. The current model, based on historical cohort survival rates and averages for entry grades, consistently overestimates school enrollment, leading to budget inefficiencies and suboptimal resource allocation. A more accurate, data-driven ML model will improve financial management by ensuring schools are appropriately funded before the start of the school year. This will free up resources, support capital planning, and optimize staffing for various student groups, including special education and bilingual students.

Problem Statement and Current Challenges

The district's existing projection model faces several key challenges:

- **Inaccurate Entry Grade Projections:** Relying on historical averages for entry grades (PK, K, 9th) leads to overestimations, especially in a district with declining enrollment.
- **Lack of Granularity:** The current process struggles to provide accurate projections for small schools due to high variability. It also cannot project student demographics, such as race/ethnicity, special education status, and English learner status, which are critical for targeted staffing and resource allocation.
- **Limited Data Inputs:** The current model does not account for external factors like population changes, new housing developments, or the impact of school choice and specialized programs. This limits its predictive power.
- **Budgetary Inefficiency:** Over-projection results in over-budgeting for schools, as funding is not reduced for those with lower-than-projected enrollment. Once funding is allocated to schools, it has been past practice to hold those schools harmless, so as not to disrupt staffing and scheduling once the school year has begun.

Proposed Machine Learning Solution

We propose developing a machine learning model to predict school-level and grade-level enrollment and zoned school assignment, including student demographic breakdowns. The model will move beyond simple historical averages by incorporating a wide range of relevant features to capture the complex dynamics of student enrollment in a choice-based district.

Project Outcomes and Benefits

Implementing this ML model will provide several benefits to the school district:

- **Improved Budgetary Accuracy:** More precise enrollment projections will lead to more accurate spring budget allocations, reducing overallocations and ensuring resources are used efficiently.
- **Optimized Staffing:** Projections of special education and EL student numbers will allow for proactive and appropriate staffing, ensuring students receive the support they need from the first day of school.
- **Enhanced Capital Planning:** Accurate long-term projections will provide a solid basis for decisions about school expansions, school boundary changes, consolidations, or closures.
- **GoCPS Seat Allocation:** Enhances the allocation of school seats used for school offers, preventing schools from overenrollment.
- **Data-Driven Decision Making:** The model's insights will help the district understand the key drivers of enrollment changes, empowering leadership to make more informed strategic decisions.

Appendix: Data and Feature Engineering

The success of the model hinges on a comprehensive dataset. While this is not an extensive list, the following features can be used as inputs for the ML model:

- **Historical Enrollment Data:** Longitudinal data on enrollment by school, grade, and student demographics (race/ethnicity, special education, bilingual status).
- **Geospatial and Zoned School Data:**
 - **Zoned School Assignment:** A student's assigned zoned school, even if they attend a different one.
 - **Residential Data:** The student's home address, enabling analysis of enrollment patterns in relation to their zoned school.
 - **Geographical Features:** School location (latitude/longitude) and proximity to major population centers, public transportation, or commercial areas.
- **Demographic and Socioeconomic Data:**
 - **Census Data:** Population changes at the census block level, including changes in the number of children in different age groups.
 - **Language Data:** Changes in the prevalence of different languages spoken in specific residential areas.
 - **Socioeconomic Indicators:** Data on income levels or other socioeconomic factors that may correlate with school choice.
- **Real Estate and Development Data:**
 - **New Construction:** The number and type of residential housing permits issued within a school's attendance zone. This can be a leading indicator of future enrollment.
 - **Commercial/Industrial Development:** Information on new businesses or industrial projects that may attract families to an area or alter the residential landscape.
- **School-Specific Data:**
 - **School Closures/Consolidations:** The impact of recent closures or consolidations on enrollment trends in neighboring schools.
 - **Application Data:** For choice schools, historical data on the number of applications received and the acceptance rates.
 - **Enrollment Trends:** Month-to-month enrollment changes within a school year to capture patterns of student mobility.